MRV: Interpreting Medical Reports into Voice in Regional Languages

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Abstract

Understanding medical reports is difficult because of complex terminologies used in medical reports. This study proposes the creation of a healthcare application using AI for interpreting the medical reports and making the reports more user-friendly. Medical reports such as CBC, thyroid, and Pulse Reports can be uploaded by the users into the app and using Optical Character Recognition (OCR), data in the report can be extracted. The Large Language Model (LLM) uses this information to interpret the medical report in simple language. Explanation of this interpretation is given in the form of both text and audio in regional languages using Text-to-Speech (TTS) Technology. In addition, the app provides pictorial explanation of body parts involved in the medical condition, important terms in the medical field, and tutorial videos for the first-time users. In addition to these features, Report History allows the user to view past reports while Health Trend Analysis allows the user to track health parameters.

Keywords: AI, OCR, LLM, TTS, Medical Report Interpretation, Regional Languages, Report History, Health Trend Analysis.

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